Adam-type stochastic approximation under controlled Markov noise

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1 Introduction

2 Controlled Markov stochastic approximation

2.1 Controlled observations and the slow recursion

Let $\Theta \subseteq \mathbb{R}^d$ carry its Borel sigma-algebra, and let $(\mathsf{X}, \mathcal{X})$ be a standard Borel state space, for example a Polish space with its Borel sigma-algebra. For each $\theta \in \Theta$, let P_θ be a Markov transition kernel on X, with $(\theta, x) \mapsto P_\theta(x, A)$ measurable for every $A \in \mathcal{X}$. Let $H : \Theta \times \mathsf{X} \to \mathbb{R}^d$ be measurable and write $H_\theta(x) := H(\theta, x)$.

On a filtered probability space, take (θ_n, X_n) to be $\mathcal{F}_n$-measurable and impose the controlled Markov condition

$$X_{n+1} \mid \mathcal{F}_n \sim P_{\theta_n}(X_n, \cdot)\,, \qquad n \geq 0. \tag{2.1}$$

For a measurable function f, the frozen kernel also acts as an operator:

$$P_\theta f(x) := \int_{\mathsf{X}} f(y)\, P_\theta(x, \mathrm{d}y)\,. \tag{2.2}$$

Equivalently, Equation 2.1 says that

$$\mathbb{E}[f(X_{n+1}) \mid \mathcal{F}_n] = P_{\theta_n} f(X_n) \tag{2.3}$$

for every bounded measurable f. The same identity holds for nonnegative f, allowing extended values, and for integrable $f(X_{n+1})$. For vector-valued functions we apply the operator componentwise.

Given deterministic positive gains $(\gamma_n)_{n \geq 1}$, the basic stochastic approximation recursion is

$$\theta_{n+1} = \theta_n + \gamma_{n+1} H_{\theta_n}(X_{n+1}). \tag{2.4}$$

The gain γ_{n+1} sets the slow parameter scale, while P_{θ_n} generates the next observation. Thus the parameter both determines the observation law and is updated from that observation: the feedback is $\theta_n \longmapsto P_{\theta_n} \longmapsto X_{n+1} \longmapsto \theta_{n+1}$. The recursion is required to remain in Θ; ensuring this and proving stability, or introducing a stabilisation mechanism, are separate tasks.

2.2 Frozen invariant marginals and the ordinary mean field

Fixing θ gives a homogeneous Markov chain with transition kernel P_θ. By contrast, Equation 2.1 uses the changing, history-dependent parameter θ_n; the observations alone need not form a Markov chain on X.

Assumption 2.1 (Frozen invariant marginal and integrability)**.** *For each $\theta \in \Theta$, choose an invariant probability π_θ for P_θ, and suppose*

$$\pi_\theta P_\theta = \pi_\theta, \qquad \int_{\mathsf{X}} \|H_\theta(x)\|\, \pi_\theta(\mathrm{d}x) < \infty.$$

Here invariance means $\pi_\theta(P_\theta f) = \pi_\theta(f)$ for every bounded measurable f. This assumption supplies a reference marginal, not a mixing rate or uniqueness. If several invariant probabilities exist, the field below depends on the specified choice. Measurability and regularity of that choice as θ varies are additional questions.

Under Assumption 2.1, define the ordinary frozen mean field by

$$h(\theta) := \pi_\theta(H_\theta) = \int_{\mathsf{X}} H_\theta(x)\, \pi_\theta(\mathrm{d}x). \tag{2.5}$$

The centred observable is $\bar{H}_\theta(x) := H_\theta(x) - h(\theta)$. Consequently $\pi_\theta(\bar{H}_\theta) = 0$, but $\bar{H}_{\theta_n}(X_{n+1})$ need not have conditional mean zero given $\mathcal{F}_n$: the conditional law in Equation 2.1 is $P_{\theta_n}(X_n, \cdot)$, not π_{θ_n}.

Remark 2.1 (A common target in adaptive MCMC). In the adaptive-MCMC special case one has $\pi_\theta \equiv \pi$, so each frozen kernel preserves the same target π. This alone does not make the adaptive observations stationary at finite times. In particular, when the kernel choice depends on the observed history, common invariance does not by itself propagate the marginal π through that choice, even if X_0 has law π.

Classical controlled Markov SA seeks conditions under which the weighted accumulation of $\bar{H}_{\theta_n}(X_{n+1})$ becomes negligible over bounded intervals of SA time. Together with stability and suitable regularity, this connects the slow recursion to the ODE

$$\dot{\theta} = h(\theta). \tag{2.6}$$

No such approximation or convergence theorem is asserted here.

2.3 The frozen stationary path law

For fixed θ, the stationary finite-dimensional distributions specified by π_θ and P_θ extend on this standard Borel space to a two-sided stationary Markov chain $(X_k^\theta)_{k\in\mathbb{Z}}$. Denote its law on $(\mathsf{X}^{\mathbb{Z}}, \mathcal{X}^{\otimes\mathbb{Z}})$ by $\mathbb{P}_\theta^{\mathrm{stat}}$. Its one-time marginal is π_θ and its forward transition kernel, conditional on the entire past, is P_θ.

These two probability laws have different domains: π_θ describes one state, whereas $\mathbb{P}_\theta^{\mathrm{stat}}$ describes an entire trajectory and therefore its temporal dependence. The path law is not in general a product law. For an ordinary one-step observable only its marginal matters:

$$\mathbb{E}_{\mathbb{P}_\theta^{\mathrm{stat}}}\big[H_\theta(X_0^\theta)\big] = \pi_\theta(H_\theta) = h(\theta). \tag{2.7}$$

History-dependent variables will instead be defined as functions of the stationary path. Expectations under $\mathbb{P}_\theta^{\mathrm{stat}}$ always refer to this frozen construction, not to the law of the adaptive observations.

2.4 Poisson equation notation

For a measurable function $g_\theta : \mathsf{X} \to \mathbb{R}^d$, the frozen Poisson equation is

$$g_\theta - P_\theta g_\theta = H_\theta - h(\theta) = \bar{H}_\theta. \tag{2.8}$$

The operator is that of Equation 2.2. To indicate its role, suppose a jointly measurable solution family is available, with the identities valid on the visited states and the required integrability. Then Equation 2.3 gives a martingale difference $\Delta_{n+1} := g_{\theta_n}(X_{n+1}) - (P_{\theta_n} g_{\theta_n})(X_n)$. Substitution into Equation 2.8 yields

$$\bar{H}_{\theta_n}(X_{n+1}) = \Delta_{n+1} + (P_{\theta_n} g_{\theta_n})(X_n) - (P_{\theta_n} g_{\theta_n})(X_{n+1}).$$

After multiplication by gains and summation, the last two terms give boundary, gain-variation, and parameter-change remainders by summation by parts. Existence of g_θ, its growth bounds, and its regularity in θ are later hypotheses or proof obligations. Invariance alone does not provide these properties or control the remainders.

2.5 SA time and the ODE language

For the ODE description take $\gamma_n \to 0$ and $\sum_{n\geq 1} \gamma_n = \infty$, in addition to positivity. Set

$$\begin{aligned} t_0 = 0, \qquad t_n &= \sum_{k=1}^{n} \gamma_k, \\ \bar{\theta}(t) = \theta_n + \frac{t - t_n}{\gamma_{n+1}}(\theta_{n+1} - \theta_n), \qquad & t \in [t_n, t_{n+1}]. \end{aligned} \tag{2.9}$$

For a general Θ, the straight segments in Equation 2.9 need not remain in Θ. Applying the following definition requires a domain containing the interpolated path on which the ODE has a continuous semiflow.

Definition 2.1 (Asymptotic pseudo-trajectory)**.** Suppose Equation 2.6 generates a continuous semiflow $\Phi : [0,\infty) \times D \to D$ on a domain $D \subseteq \Theta$. A continuous path $\bar{\theta} : [0,\infty) \to D$ is an *asymptotic pseudo-trajectory* (APT) of Φ if, for every $T > 0$,

$$\lim_{t\to\infty} \sup_{0\leq s\leq T} \left\| \bar{\theta}(t+s) - \Phi_s(\bar{\theta}(t)) \right\| = 0.$$

Thus each late finite-horizon segment follows the ODE started from its own initial point. Well-posedness of the ODE, stability of the interpolation, and the noise estimates needed for this property remain to be verified; the definition itself supplies none of them.

Ordinary controlled Markov SA therefore has the one-step frozen mean field $h(\theta) = \pi_\theta(H_\theta)$ of Equation 2.5. The next section asks what happens when the update is first passed through a stable linear memory.

3 Momentum and preservation of the mean field

3.1 Fixed linear memory

In the setup of Section 2, fix $0 \le \alpha < 1$ and introduce an $\mathbb{R}^d$-valued momentum variable with M_0 measurable with respect to $\mathcal{F}_0$. Replace Equation 2.4 by

$$M_{n+1} = \alpha M_n + (1-\alpha) H_{\theta_n}(X_{n+1}) \tag{3.1}$$

and

$$\theta_{n+1} = \theta_n + \gamma_{n+1} M_{n+1}. \tag{3.2}$$

The filtration includes the momentum history, and the observations still satisfy Equation 2.1. The coefficient α is fixed; it is not another vanishing SA gain. At $\alpha = 0$ the recursion reduces to the ordinary update.

3.2 Stationary frozen momentum

Fix θ and use the two-sided stationary frozen chain with path law $\mathbb{P}_\theta^{\mathrm{stat}}$ from Section 2. Define its stationary momentum by

$$M^\theta := (1-\alpha) \sum_{j=0}^{\infty} \alpha^j H_\theta(X_{-j}^\theta). \tag{3.3}$$

At $\alpha = 0$ this expression means $M^\theta = H_\theta(X_0^\theta)$.

Lemma 3.1 (Integrable stationary memory)**.** *Fix θ. If the frozen observations are stationary with marginal π_θ and $\pi_\theta(\|H_\theta\|) < \infty$, the series in Equation 3.3 converges absolutely almost surely and in L^1 under $\mathbb{P}_\theta^{\mathrm{stat}}$.*

Proof. Put $c_\theta := \int \|H_\theta(x)\| \, \pi_\theta(\mathrm{d}x) < \infty$. Tonelli's theorem and stationarity give

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_\theta^{\mathrm{stat}}} \left[(1-\alpha) \sum_{j=0}^{\infty} \alpha^j \left\| H_\theta(X_{-j}^\theta) \right\| \right] &= (1-\alpha) \sum_{j=0}^{\infty} \alpha^j c_\theta \\ &= c_\theta. \end{aligned}$$

The nonnegative series of norms is consequently finite almost surely. If S_J denotes the partial sum in Equation 3.3 through $j = J$, the same bound on its tail gives $\mathbb{E}_{\mathbb{P}_\theta^{\mathrm{stat}}}[\|M^\theta - S_J\|] \le c_\theta \alpha^{J+1} \to 0$. This proves the L^1 assertion as well. □

3.3 Preservation of the frozen field

Proposition 3.1 (Momentum preserves the frozen mean field)**.** *For fixed θ and $0 \le \alpha < 1$, suppose the frozen Markov chain with kernel P_θ is stationary with marginal π_θ and $\pi_\theta(\|H_\theta\|) < \infty$. Then the stationary momentum in Equation 3.3 satisfies*

$$\mathbb{E}_{\mathbb{P}_\theta^{\mathrm{stat}}}[M^\theta] = h(\theta). \tag{3.4}$$

Proof. The integrable series-of-norms bound in Lemma 3.1 justifies interchanging the sum and expectation, componentwise. Stationarity and Equation 2.7 then yield

$$\begin{aligned}\mathbb{E}_{\mathbb{P}_\theta^{\mathrm{stat}}}\left[M^\theta\right] &= (1-\alpha)\sum_{j=0}^{\infty}\alpha^j\mathbb{E}_{\mathbb{P}_\theta^{\mathrm{stat}}}\left[H_\theta(X_{-j}^\theta)\right]\\ &= (1-\alpha)\sum_{j=0}^{\infty}\alpha^j h(\theta)\\ &= h(\theta).\end{aligned}$$

No independence across time is used. The calculation uses only stationarity, integrability, and the normalisation $(1-\alpha)\sum_{j\geq 0}\alpha^j = 1$; indeed, this field identity holds for any stationary observation sequence with the same marginal and first-moment bound. □

The equality in Equation 3.4 is stronger than equality of roots: it leaves the entire frozen first-order Robbins–Monro field unchanged. In particular,

$$\mathbb{E}_{\mathbb{P}_\theta^{\mathrm{stat}}}\left[M^\theta\right] = 0 \quad\Longleftrightarrow\quad h(\theta) = 0.$$

This is a stationary field identification; it does not by itself prove convergence of the adaptive recursion Equation 3.2.

Remark 3.1 (Initialisation in the frozen system). For frozen stationary observations and any integrable initial M_0, iteration of the frozen recursion gives, for $n \geq 1$,

$$M_n = \alpha^n M_0 + (1-\alpha)\sum_{j=0}^{n-1}\alpha^j H_\theta(X_{n-j}^\theta). \tag{3.5}$$

Taking expectations and summing the finite geometric series gives

$$\mathbb{E}[M_n] = \alpha^n\mathbb{E}[M_0] + (1-\alpha^n)h(\theta). \tag{3.6}$$

Thus $M_0 = 0$ gives $\mathbb{E}[M_n] = (1-\alpha^n)h(\theta)$, and the initialisation bias relative to $h(\theta)$ decays geometrically. No independence between M_0 and the observations is needed for this expectation calculation. These are frozen identities; they do not replace θ_n by a constant in the actual adaptive recursion.

3.4 Discussion point: augmented-state lifting

The direct argument of Proposition 3.1 already establishes preservation of the entire frozen Robbins–Monro field without enlarging the state. A stronger alternative interpretation is to adjoin the memory state, $Y_n = (X_n, M_n)$, representing momentum as a controlled Markov stochastic approximation with a lifted kernel Q_θ. For discussion with Matt and Sam, the potentially reusable question is whether suitable assumptions on the base frozen family P_θ can be transferred to Q_θ strongly enough to invoke existing controlled-Markov SA machinery, through invariant-law, coupling or mixing, Poisson-equation, and parameter-regularity estimates. This possible lifting principle and the required transfer estimates are not pursued here.

4 The stationary Adam field

4.1 Nonlinear fading memory

In Section 3, linear memory preserved the entire frozen field: $\mathbb{E}_{\mathbb{P}_\theta^{\mathrm{stat}}}[M^\theta] = h(\theta)$. Adam adds a second-moment memory and divides the momentum by the regularised square root of that memory. Fix $0 \le \alpha, \beta < 1$ and $\varepsilon > 0$, and consider the controlled observations of Equation 2.1 with

$$M_{n+1} = \alpha M_n + (1-\alpha) H_{\theta_n}(X_{n+1}), \tag{4.1}$$

$$V_{n+1} = \beta V_n + (1-\beta) H_{\theta_n}(X_{n+1})^{\odot 2}, \tag{4.2}$$

$$\theta_{n+1} = \theta_n + \gamma_{n+1} \frac{M_{n+1}}{\sqrt{V_{n+1}} + \varepsilon}. \tag{4.3}$$

Squares, square roots, division, and the product $\odot$ are componentwise; the scalar ε is added in each coordinate. Initially take $M_0 \in \mathbb{R}^d$ and $V_0 \in [0,\infty)^d$, measurable with respect to $\mathcal{F}_0$. These are uncorrected memories, with no finite-time bias factors. We view this as an SA mechanism, without requiring H to be a gradient. The two memories are linear filters of the signal and its square, but their data-dependent normalisation is nonlinear: this is where the mean-field preservation argument of Proposition 3.1 ceases to apply.

4.2 Frozen memories and the candidate drift

Fix θ and work under the stationary path law $\mathbb{P}_\theta^{\mathrm{stat}}$ of Section 2. Throughout the frozen construction assume $\pi_\theta(\|H_\theta\|^2) < \infty$. Alongside M^θ from Equation 3.3, define

$$V^\theta := (1-\beta) \sum_{j=0}^{\infty} \beta^j H_\theta(X_{-j}^\theta)^{\odot 2}. \tag{4.4}$$

When a memory coefficient is zero, its series means just its $j = 0$ term. For each coordinate i, Tonelli's theorem and stationarity give

$$\mathbb{E}_{\mathbb{P}_\theta^{\mathrm{stat}}}[V_i^\theta] = (1-\beta) \sum_{j \ge 0} \beta^j \pi_\theta(H_{\theta,i}^2) = \pi_\theta(H_{\theta,i}^2) < \infty.$$

Thus the nonnegative series is finite almost surely and integrable; its partial sums converge in L^1 as well, since their expected tails are $\beta^{J+1}\pi_\theta(H_{\theta,i}^2)$ after index J. The same moment assumption implies the hypothesis of Lemma 3.1, so M^θ exists in L^1. Neither argument uses independence.

Definition 4.1 (Stationary Adam field)**.** Under these frozen moment assumptions, put $W^\theta := (\sqrt{V^\theta} + \varepsilon)^{-1}$ and $A^\theta := M^\theta \odot W^\theta$. The *stationary Adam field* is

$$F(\theta) := \mathbb{E}_{\mathbb{P}_\theta^{\mathrm{stat}}}[A^\theta] = \mathbb{E}_{\mathbb{P}_\theta^{\mathrm{stat}}}\left[\frac{M^\theta}{\sqrt{V^\theta} + \varepsilon}\right]. \tag{4.5}$$

This expectation exists because $0 < W_i^\theta \le \varepsilon^{-1}$ and $\|A^\theta\| \le \varepsilon^{-1}\|M^\theta\|$. Unlike $h(\theta) = \pi_\theta(H_\theta)$, which uses only a one-time marginal, $F(\theta)$ uses the stationary path law through both fading histories. It is a candidate frozen drift, not yet an ODE approximation for the actual adaptive recursion Equation 4.3. Regularity in θ is not a consequence of this pointwise definition.

4.3 Relation to the IID Adam field

[1, Definition 1.3] define the corresponding field by averaging the normalised momentum computed from an infinite IID history. Their Lemma 3.1 gives a bounded, 1-Lipschitz history map in a specific weighted sequence norm; with their locally p-regular innovation (Definition 1.4), the field is bounded and locally Lipschitz when $\alpha < \sqrt{\beta}$. That lemma recalls earlier work by Dereich and Jentzen, rather than proving a new history-map estimate.[1]

For IID innovations, their Theorem 1.5 gives localised finite-horizon ODE tracking under local p-regularity, $p \ge 2$, and positive, non-summable, decreasing gains tending to zero with their step-size regularity condition (10). With $\sum_n \gamma_n^{1+p/3} < \infty$, Corollary 1.9 gives the APT property on the event of bounded iterates. Corollary 1.10 requires any limit in probability to be a zero of the Adam field. As discussed after Definition 1.4, that field can have different zeros and directions from the ordinary mean field [1]. Their Definition 1.1 uses a finite-time second-moment bias correction; the stationary history functional agrees with ours, but the finite recursion differs. Neither their IID theorem nor that agreement proves the controlled-Markov APT result sought here.

4.4 A pathwise bound on the normalised momentum

The boundedness part of the cited history-map estimate has a short proof that does not involve a probability law.

Lemma 4.1 (Uniform bound for normalised histories)**.** *Suppose $0 \le \alpha < \sqrt{\beta} < 1$. For a scalar history $(z_j)_{j\ge 0}$ with $\sum_j \beta^j z_j^2 < \infty$, set*

$$m = (1-\alpha)\sum_{j\ge 0}\alpha^j z_j, \qquad v = (1-\beta)\sum_{j\ge 0}\beta^j z_j^2.$$

Then the numerator converges absolutely and

$$\frac{|m|}{\sqrt{v}+\varepsilon} \le c_{\alpha,\beta} := \frac{1-\alpha}{\sqrt{1-\beta}\sqrt{1-\alpha^2/\beta}}. \tag{4.6}$$

Applying this coordinatewise to vector histories gives the Euclidean bound $\|A\| \le C_{\alpha,\beta,d} := \sqrt{d}\, c_{\alpha,\beta}$.

Proof. Weighted Cauchy–Schwarz, first for partial sums and then for the nonnegative series of absolute values, gives

$$\begin{aligned}(1-\alpha)\sum_{j\ge 0}\alpha^j\,|z_j| &\le (1-\alpha)\left[\sum_{j\ge 0}(\alpha^2/\beta)^j\right]^{1/2}\left[\sum_{j\ge 0}\beta^j z_j^2\right]^{1/2}\\ &= c_{\alpha,\beta}\sqrt{v}.\end{aligned}$$

[1] [2], Lemma 3.1 and equation (148). The former proves the deterministic history-map bound and Lipschitz estimate; the latter bounds the induced field's parameter variation. These are the results explicitly recalled in Lemma 3.1 of [1].

Division by $\sqrt{v}+\varepsilon$ proves Equation 4.6, including $v=0$. Summing squared coordinate bounds proves the vector assertion. □

This estimate is deterministic and pathwise; it applies to stationary Markov histories just as to IID histories. It bounds $F(\theta)$ uniformly wherever the frozen moment assumption holds, but supplies no parameter regularity or controlled-Markov ODE theorem.

4.5 Bounded velocity and adaptive-MCMC validity

For the actual recursion, now take $M_0 = V_0 = 0$ and $\alpha < \sqrt{\beta}$. Unrolling both memories gives a finite history of the actual signals $H_{\theta_{k-1}}(X_k)$, padded with zeros. Lemma 4.1 therefore yields, without freezing parameters,

$$\|\theta_{n+1} - \theta_n\| \le C_{\alpha,\beta,d}\gamma_{n+1}. \tag{4.7}$$

This uniform constant does not cover arbitrary initial memories without an additional compatibility bound. Bounded velocity is not parameter convergence.

Suppose also that $\gamma_n \to 0$, $\pi_\theta \equiv \pi$, the parameters remain in the admissible set, and the frozen family satisfies

$$\sup_x d_{\mathrm{TV}}(P_\theta(x,\cdot), P_\vartheta(x,\cdot)) \le L\,\|\theta-\vartheta\|, \qquad \theta,\vartheta\in\Theta. \tag{4.8}$$

Here $d_{\mathrm{TV}}(\mu,\nu) := \sup_B |\mu(B)-\nu(B)|$. This is assumption (A8) of [3]. In particular,

$$D_n := \sup_x d_{\mathrm{TV}}(P_{\theta_n}(x,\cdot), P_{\theta_{n-1}}(x,\cdot)) \le LC_{\alpha,\beta,d}\gamma_n \longrightarrow 0. \tag{4.9}$$

For measurability one may use the deterministic upper bound $\min\{1, LC_{\alpha,\beta,d}\gamma_n\}$ in place of the supremum. This gives their diminishing adaptation (A6), in fact pathwise. Their Lemma 10 implies weak 1-waning; here Cesàro averaging also directly gives $n^{-1}\sum_{k=1}^n D_k \to 0$ almost surely, their *strong* 1-*waning* condition (A5). Lemma 16 treats the general SA-sized-increment mechanism [3].

An LLN needs more. Under their adapted controlled-transition setup (A1), the common target, and simultaneous uniform ergodicity (A3),

$$\sup_{\theta\in\Theta}\sup_x d_{\mathrm{TV}}(P_\theta^k(x,\cdot),\pi) \le C\rho^k, \qquad k\ge 0, \quad C<\infty, \quad 0\le\rho<1,$$

Theorem 9(ii) then gives $n^{-1}\sum_{k=1}^n f(X_k) \to \pi(f)$ almost surely for every fixed bounded measurable f [3]. Neither common invariance nor diminishing adaptation alone establishes this conclusion. Keeping θ_n in the admissible set is essential; their Remark 15 discusses stabilisation, which requires specifying the actual algorithm. This sampling-validity conclusion does not require parameter convergence and does not identify its equilibria: it can coexist with nonconvergent adaptation or an Adam equilibrium different from a Robbins–Monro root.

4.6 Covariance and Robbins–Monro roots

Return to the frozen moment assumptions of Definition 4.1; the following identities require no restriction $\alpha < \sqrt{\beta}$. All expectations and covariances in this subsection are under $\mathbb{P}_\theta^{\mathrm{stat}}$. Since M^θ is integrable and W^θ is bounded, all coordinate products below are integrable. By Proposition 3.1,

$$F_i(\theta) = h_i(\theta)\mathbb{E}[W_i^\theta] + \mathrm{Cov}(M_i^\theta, W_i^\theta).$$

Define the *componentwise covariance vector*, not a covariance matrix, by $\mathrm{Cov}_\odot(M, W)_i := \mathrm{Cov}(M_i, W_i)$. Thus

$$F(\theta) = h(\theta) \odot \mathbb{E}[W^\theta] + \mathrm{Cov}_\odot(M^\theta, W^\theta). \tag{4.10}$$

This is exact: expand each covariance as $\mathbb{E}[M_i^\theta W_i^\theta] - \mathbb{E}[M_i^\theta]\mathbb{E}[W_i^\theta]$ and use $\mathbb{E}[M_i^\theta] = h_i(\theta)$.

Proposition 4.1 (A covariance criterion at a Robbins–Monro root)**.** *Let $\theta^\star$ satisfy the frozen moment assumptions and $h(\theta^\star) = 0$. Then*

$$F(\theta^\star) = \mathrm{Cov}_\odot(M^{\theta^\star}, W^{\theta^\star}). \tag{4.11}$$

Consequently $\theta^\star$ is also an Adam root if and only if $\mathrm{Cov}(M_i^{\theta^\star}, W_i^{\theta^\star}) = 0$ for every coordinate.

Proof. Substitute $h(\theta^\star) = 0$ in Equation 4.10. The vector vanishes exactly when all its coordinates vanish. □

A zero stationary numerator mean is therefore insufficient: the random normalisation uses the same history, and root correspondence requires additional structure in the joint law. Conversely, an Adam root with $h(\theta) \neq 0$ would satisfy the cancellation identity $h(\theta) \odot \mathbb{E}[W^\theta] = -\mathrm{Cov}_\odot(M^\theta, W^\theta)$. There is no general reason for the two root sets to agree in either direction; this is not a claim that they differ in every model.

Both memories use $(\ldots, X_{-1}^\theta, X_0^\theta)$. Even IID observations can create numerator–normaliser dependence through shared inputs; Markov observations introduce additional lagged dependence. Two frozen kernels with the same invariant marginal can therefore, in principle, give different Adam fields through their different stationary path laws. No such counterexample is asserted here.

The memory existence, field integrability, pathwise bound, and covariance criterion above are direct derivations. Controlled-Markov Adam APT convergence, parameter regularity of F, existence or uniqueness of equilibria, parameter convergence, and any augmented-state transfer of mixing or Poisson estimates remain unproved. Section 5 takes up the passage from IID histories to controlled Markov histories.

5 From IID histories to controlled Markov histories

6 Main result and proof architecture

7 Adaptive MCMC consequences

8 Example / numerical illustration

9 Discussion and general moment-augmentation interface

References

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